Formal Proof: The Torsional Trefoil Markov Pump Theorem (TTMPT)

Abstract

The Torsional Trefoil Markov Pump Theorem (TTMPT) establishes a deterministic mathematical framework for a one-way stochastic pump mechanism acting on a longitudinal binding-scalar field. This paper formalizes the 10-step macro-cycle operating over a 12-channel lattice Hilbert space ($\mathcal{H}_{\Lambda_{12}}$). By introducing dynamic transverse variables (π^2 and $\sqrt{\pi}$) as orthogonal heat sinks, formally postulating the projection of Hilbert-space amplitudes into classical Markov probabilities, and utilizing a topological "Tesla-valve" Markovian kernel, we rigorously prove that the system breaks detailed balance. Utilizing explicit stochastic boundary conditions, we demonstrate that the TTMPT generates a strictly positive, continuous net-forward probability current, advancing the binding scalar field while resetting the modular rotational phase to identity.

1. Axiomatic Foundations and State Space

1.1 The Rigorous State Space

The structural bedrock of the system consists of a triple-trefoil skeleton. We rigorously define the localized state space as a 12-channel lattice Hilbert space:

$$\mathcal{H}_{\Lambda_{12}} \cong \mathbb{C}^{12}$$

The space operates using an explicit basis representing three coupled 4-phase streams (denoted by phase groups α, β, γ). A state vector at macro-step n is defined as the superposition:

$$|\psi_n\rangle = \sum_{g \in \{\alpha,\beta,\gamma\}} \sum_{r=1}^{4} c_{g,r}^{(n)} e_{g,r}$$

This state vector is subject to the standard normalization constraint ensuring conservation of total probability amplitude:

$$\sum_{g,r} |c_{g,r}^{(n)}|^2 = 1$$

1.2 Dynamical Transverse Variables

To govern the continuous spatial scaling and continuous wave energy orthogonal to the central longitudinal binding field B, we introduce a scalar deformation parameter δ alongside continuous topological perturbation modes $\hat{u}(t)$ and $\hat{v}(t)$. The transverse orthogonal foliation variables are rigorously defined as:

$$u(t) = \pi^2 + \delta\hat{u}(t) \quad \text{(Outer Manifold Variance)}$$

$$v(t) = \sqrt{\pi} + \delta\hat{v}(t) \quad \text{(Inner Skeleton Winding Parameter)}$$

These continuously parameterized variables dictate the continuous deformation of the Clifford-torus during the stochastic pump cycle. They explicitly define the structural topological resistance function $A_{ij}(t) = A_{ij}(u(t), v(t)) > 0$ for each graph edge, acting as independent orthogonal heat sinks for reverse-propagating current.

2. Topological Asymmetry and Broken Detailed Balance

2.1 The Postulated Hilbert-to-Markov Probability Bridge

To connect the quantum-like state vectors of $\mathcal{H}_{\Lambda_{12}}$ to classical stochastic dynamics, we explicitly define the projection mapping.

Definition (Classical Markov Projection). The classical Markov projection of the Hilbert-space state is the probability vector $p^{(n)}$ with entries:

$$p_{g,r}^{(n)} = |c_{g,r}^{(n)}|^2$$

Consolidating these probabilities yields a 12-dimensional classical probability vector $p^{(n)} = (p_1^{(n)}, \ldots, p_{12}^{(n)})$. We postulate that, under the coarse-grained pump dynamics, this projected probability vector evolves according to a row-stochastic Markov kernel K such that:

$$p^{(n+1)} = p^{(n)}K$$

2.2 Broken Detailed Balance and the Asymmetric Kernel

For a finite Markov chain with stationary distribution π, detailed balance holds if $\pi_i K_{ij} = \pi_j K_{ji}$ for all states i, j. To break detailed balance, we define a non-empty oriented forward edge set $E^+ \neq \varnothing$.

Lemma 2.1 (Asymmetric Markovian Kernel). Suppose that for every $(i, j) \in E^+$, the row-stochastic transition probabilities satisfy:

$$K_{ij} > 0, \qquad K_{ji} = K_{ij} e^{-\lambda A_{ij}}$$

where $\lambda > 0$, and $A_{ij} > 0$. If additionally, the stationary distribution satisfies the structural condition:

$$\pi_i > \pi_j e^{-\lambda A_{ij}}$$

then it strictly follows that:

$$\pi_i K_{ij} - \pi_j K_{ji} > 0$$

Hence, the net probability current $J = \sum_{(i,j) \in E^+} (\pi_i K_{ij} - \pi_j K_{ji})$ over the non-empty set is strictly positive.

Proof. The current along any directed edge (i, j) is $J_{ij} = \pi_i K_{ij} - \pi_j K_{ji}$. Substituting the topological suppression rule $K_{ji} = K_{ij} e^{-\lambda A_{ij}}$ gives:

$$J_{ij} = K_{ij} \left(\pi_i - \pi_j e^{-\lambda A_{ij}} \right)$$

Because we have explicitly assumed $K_{ij} > 0$ and $\pi_i > \pi_j e^{-\lambda A_{ij}}$, the term in parentheses is strictly positive. Therefore, $J_{ij} > 0$ for every edge in E^+. Since $E^+ \neq \varnothing$, there exists at least one edge (i, j) such that $\pi_i K_{ij} \neq \pi_j K_{ji}$, so detailed balance fails. Consequently, the net current J over the non-empty set is the sum of strictly positive terms.

3. Mechanics of the 10-Step Macro-Cycle

3.1 Phase I: Compression and the Frame 5 Solver

Let the 12 -channel state lie in a Hilbert space $\mathcal{H} \cong \mathbb{C}^{12}$, and assume $\||\psi_0\rangle\| = 1$. Let $P_1, \ldots, P_4$ be orthogonal projections on $\mathcal{H}$ and let Π be a unitary discrete phase operator. Define the compression operator:

$$C = P_4 P_3 \Pi P_2 P_1$$

Since each P_k is norm non-increasing and Π is norm-preserving, C is norm non-increasing ($\|C|\psi\rangle\| \leq \||\psi\rangle\|$). Assuming $C|\psi_0\rangle \neq 0$, define the normalized pre-pump compressed state by:

$$|\psi_{\text{comp}}\rangle = \frac{C|\psi_0\rangle}{\|C|\psi_0\rangle\|} = \frac{P_4 P_3 \Pi P_2 P_1 |\psi_0\rangle}{\|P_4 P_3 \Pi P_2 P_1 |\psi_0\rangle\|}$$

Then $\||\psi_{\text{comp}}\rangle\| = 1$. The Frame 5 Algebraic Solver provides the geometric phase-lock parameter $s = 1 - \frac{(Q_\alpha + Q_\beta - Q_\gamma)^2}{4 Q_\alpha Q_\beta}$ with $Q_\alpha, Q_\beta > 0$. Setting $s = \sin^2\left(\frac{2\pi}{3}\right) = \frac{3}{4}$ imposes the symmetric 120° phase-lock condition. This geometric condition motivates the topology of the cycle, while the rigorous contraction mechanism is supplied by the Markov kernel assumptions in the pump phase.

3.2 Phase II: Core Pump Stroke and Time Parameterization

During the pump interval $t \in [t_5, t_7]$, let $K(t)$ be a continuously parameterized family of stochastic transition matrices. For example, in the row-stochastic convention, $K_{ij}(t) \geq 0$ and $\sum_j K_{ij}(t) = 1$. The compressed state induces a probability vector:

$$p_i^{\text{comp}} = |\langle e_i | \psi_{\text{comp}} \rangle|^2$$

The Markov kernel evolves this probability layer during the pump stroke. If the larger theorem requires strict contraction, assume there exists $0 \leq \lambda < 1$ such that $\|pK(t) - qK(t)\|_1 \leq \lambda \|p - q\|_1$ for all admissible probability vectors p, q and all

$t \in [t_5, t_7]$. Let S_1, S_2, S_3 be unitary rotational operators ($S_k^* S_k = I$, so $S_k^{-1} = S_k^*$). Assuming $\sum_{k=1}^{3}(S_k + S_k^{-1})|\psi_0\rangle \neq 0$, define the normalized quaternion centroid by:

$$|\Psi_4\rangle = \frac{\sum_{k=1}^{3}(S_k + S_k^{-1})|\psi_0\rangle}{\left\|\sum_{k=1}^{3}(S_k + S_k^{-1})|\psi_0\rangle\right\|}$$

The active current $J(t)$ is then interpreted as a parameterized flow directed toward $|\Psi_4\rangle$.

3.3 Phase III: Double Toroidal Collapse

Let $(u(t), v(t)) \in \mathbb{T}^2$ parameterize the toroidal collapse, and let $A_{ij} : \mathbb{T}^2 \to \mathbb{R}_{\geq 0}$ be continuous topological resistance functions. Introduce a reverse-energy functional $E_{\text{rev}}(t) \geq 0$. The collapse phase is dissipative if:

$$\frac{d}{dt}E_{\text{rev}}(t) = -\sum_{i,j} A_{ij}(u(t), v(t)) R_{ij}(t)^2 \leq 0$$

It is strictly dissipative whenever at least one active resistance mode satisfies $A_{ij}(u(t), v(t)) R_{ij}(t)^2 > 0$. Thus reverse variance is dissipated into transverse modes while the longitudinal field remains protected by a separately imposed conservation or lower-bound condition.

3.4 Phase IV: Encapsulation and Phase Reset

Let $\Theta_k \in \mathbb{R}/2\pi\mathbb{Z}$ and define the modular phase update by $\Theta_{k+1} = \Theta_k + \omega_k$. After one full 10-step macro-cycle:

$$\Theta_{10} = \Theta_0 + \sum_{k=0}^{9} \omega_k$$

Encapsulation requires the residual modular rotation to vanish: $\sum_{k=0}^{9} \omega_k \equiv 0 \pmod{2\pi}$. Equivalently, $\Theta_{10} \equiv \Theta_0 \pmod{2\pi}$.

4. The Torsional Trefoil Markov Pump Theorem (TTMPT)

With the axiomatic mechanics established, we formalize the principal theorem.

Theorem 4.1 (TTMPT). Let Λ_{12} be a finite state space with 12 states, and let $E^+ \subseteq \Lambda_{12} \times \Lambda_{12}$ be a nonempty oriented forward edge set ($E^+ \neq \varnothing$). Let $[t_5, t_7]$ be a nondegenerate compact interval with $t_7 > t_5$. For each $t \in [t_5, t_7]$, let $K(t)$ be a continuously parameterized family of discrete-time row-stochastic, irreducible, aperiodic transition matrices. Assume that $K_{ij}(t)$, the unique stationary distribution $\pi_i(t)$, and the topological resistance function $A_{ij}(t) = A_{ij}(u(t), v(t))$ are continuous in t, satisfying:

$$\pi(t)K(t) = \pi(t), \qquad \sum_i \pi_i(t) = 1, \qquad \pi_i(t) > 0$$

Assume that for every $(i,j) \in E^+$ and every $t \in [t_5, t_7]$,

$$K_{ij}(t) > 0, \qquad K_{ji}(t) = K_{ij}(t)e^{-\lambda A_{ij}(t)}$$

where $\lambda > 0$, and $A_{ij}(t) > 0$, and assume:

$$\pi_i(t) > \pi_j(t)e^{-\lambda A_{ij}(t)}$$

Define the continuous forward probability current by:

$$J(t) = \sum_{(i,j)\in E^+} \left(\pi_i(t)K_{ij}(t) - \pi_j(t)K_{ji}(t)\right)$$

Then $J(t) > 0$ for all $t \in [t_5, t_7]$. Consequently, if B satisfies:

$$\frac{dB}{dt} = J(t)$$

then:

$$B(t_7) > B(t_5)$$

Furthermore, if the discrete phase update is $\Theta_{k+1} = \Theta_k + \omega_k$, and satisfies the closure condition $\sum_{k=0}^{9} \omega_k \equiv 0 \pmod{2\pi}$, then:

$$\Theta_{10} \equiv \Theta_0 \pmod{2\pi}$$

Proof. For each $(i,j) \in E^+$ at any time $t \in [t_5, t_7]$, using the topological suppression rule:

$$\pi_i(t)K_{ij}(t) - \pi_j(t)K_{ji}(t) = \pi_i(t)K_{ij}(t) - \pi_j(t)K_{ij}(t)e^{-\lambda A_{ij}(t)}$$

Factoring out the strictly positive $K_{ij}(t)$ yields:

$$\pi_i(t)K_{ij}(t) - \pi_j(t)K_{ji}(t) = K_{ij}(t)\left(\pi_i(t) - \pi_j(t)e^{-\lambda A_{ij}(t)}\right)$$

Since $\pi_i(t) > \pi_j(t)e^{-\lambda A_{ij}(t)}$ by hypothesis, the quantity in parentheses is strictly positive, yielding:

$$\pi_i(t)K_{ij}(t) - \pi_j(t)K_{ji}(t) > 0$$

Because E^+ is nonempty ($E^+ \neq \varnothing$), there exists at least one edge (i,j) such that $\pi_i(t)K_{ij}(t) \neq \pi_j(t)K_{ji}(t)$, so detailed balance fails at each $t \in [t_5, t_7]$. Summing over the forward edge set gives a strictly positive continuous net current:

$$J(t) = \sum_{(i,j)\in E^+} (\pi_i(t)K_{ij}(t) - \pi_j(t)K_{ji}(t)) > 0$$

Given the scalar field evolves as $\frac{dB}{dt} = J(t)$, we integrate over the non-degenerate interval $[t_5, t_7]$:

$$B(t_7) - B(t_5) = \int_{t_5}^{t_7} J(t)\,dt$$

Because $J(t) > 0$ and is continuous throughout the entire interval, and $t_7 > t_5$, the definite integral is strictly positive:

$$\int_{t_5}^{t_7} J(t)\, dt > 0$$

Hence, $B(t_7) - B(t_5) > 0$, demonstrating $B(t_7) > B(t_5)$.

Finally, iterating the discrete phase updates through the 10-step macro-cycle gives:

$$\Theta_{10} = \Theta_0 + \sum_{k=0}^{9} \omega_k$$

Using the phase-closure constraint $\sum_{k=0}^{9} \omega_k \equiv 0 \pmod{2\pi}$, we obtain:

$$\Theta_{10} \equiv \Theta_0 \pmod{2\pi}$$

Therefore, one full macro-cycle produces strictly positive advancement of the scalar field B while successfully resetting the modular phase to identity.